



OPERATING MANUAL FOR CCEA ESTUARY



The same Professor Challenger who made the Earth scream with his pain machine, as described by Arthur Conan Doyle, gave a lecture after mixing several textbooks on geology and biology in a fashion befitting his simian disposition. He explained that the Earth—the Deterritorialized, the Glacial, the giant Molecule—is a body without organs. This body without organs is permeated by unformed, unstable matters, by flows in all directions, by free intensities or nomadic singularities, by mad or transitory particles.

- Deleuze and Guattari, ATP #4 (1980)

The music of Alice Coltrane and Sun Ra, of Underground Resistance and George Russell, of Tricky and Martina, comes from the Outer Side. It alienates itself from the human; it arrives from the future. Alien Music is a synthetic recombinator, an applied art technology for amplifying the rates of becoming alien. Optimize the ratios of excentricity. Synthesize yourself.”

- Kodwo Eshun, More Brilliant Than the Sun

Futurhythmachines turn the extended capability of machines into super sensory powers.

- Kodwo Eshun, More Brilliant Than the Sun (1998)

Welcome to Estuary.

ESTUARY is a nascent hardware-software ecosystem for DIY and community embeded music making.

Recent developments have made it inexpensive to manufacture reprogrammable integrated circuits (ICs, colloquially chips) that can run sound synthesis algorithms at audio rate. What that means: we are entering an age where you can design your own sonic palettes and instruments and upload them onto mobile, screenless hardware.

At the same time, the manufacture of ICs are extremely energy-inefficient. It's not difficult to imagine this contributing to the seemingly endless faucets of e-waste moving towards our landfill and oceans. New media art must think and act permacomputationally.

CCAM developed the ESTUARY module to encourage reusability of these incredibly powerful processors, as well as the open sharing of sound synthesis algorithms between musicians and technologists.

ESTUARY is built on a microcontroller using one of these chips called Daisy Patch SM made by Electrosmith. It runs at audio rate (up to 96kHz) at high-quality bitrates. Nothing comes on the Patch SM; it needs software written in C++, Arduino, and Max/MSP (gen~).

This zine details 3 initial software profiles for the Estuary. These patches were inspired by software I designed with Juan Flores for Chicago House Music as part of CCAM's FUTURHYTHM MACHINES:: Chicago House event at Watershed Art & Ecology in May 2024 for the CCAM Earth Module. It's built using software infrastructure designed by Garnet Grimm for March 2025 workshop first showcasing the ESTUARY module.

The software on here was designed especially for participants of the Futurhythmmachines workshop at Indiana University Bloomington in the Digital Arts program. The software is available for free at <https://ccam.world>.

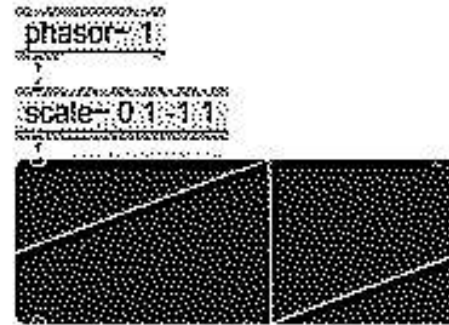
We hope to develop infrastructure for further sharing, encouraging people to share their own software for the concretized ESTUARY module. Through workshops, collaborative learning, and artistic production and presentation, we hope to build a community of artists, musicians, and technologists with different strengths, aspirations, and backgrounds.

- Grace Grace Grace, CCAM lead organizer
April 2026, Chicago IL



Control Voltage

An important principle for linking together ESTUARY modules is “Control Voltage”. By using software to program different patterns, the Daisy microcontroller can output varying voltages. ESTUARY can also receive voltages via a 3.5mm patch cable “mono” and cable). Control voltage is abbreviated as CV.



In these instruments explored in this workshop, Control Voltage is used to communicate the clock pulse or tempo, so that modules can be synchronized in time.

This is only the very beginning. Check out the visualization of the “ramp” signal to the left. Imagine we have made some machine to connect this to an automatic door; behind the door is a room with a speaker. When the signal rises, the door opens; when it falls, the door shuts again. Now consider how this would affect the sound to us in the other room; the sound would gradually become brighter and clearer as the door opened, and suddenly darker and more muffled.

What we have imaged is a low-frequency oscillator (LFO) that controls the frequency of a filter modulating a sound

source. If we imagine changing how fast the door opens and closes, we change the frequency of the LFO. If we imagine the door smoothly opening and closing, we change the waveform of the LFO. In hardware synthesis, these are questions of control voltage.

This is just one example affecting the timbre of sound. These decisions are musical ones. Changing these parameters of time creates musical dynamism that is pleasing to the ear, curious to the mind, and moving to the body. Or, not -- that's where musicianship comes in!

This "control voltage" approach to working with sound is widespread in electronic music production and performance and takes on other forms in other approaches (such as DAWs like Logic, Live, or GarageBand).

Gates

While control voltage is continuous (ranging from -12v to 12v in electric voltage or 0 to 1 in software), a GATE is another type of voltage that eurorack-format and other synthesizers use to control events. GATES are **either** LOW (close to 0) or HIGH (the highest range of voltage). You can think of GATES as a binary signal.

GATES often use dedicated hardware pins on digital microprocessors. Because they are either on or off, they're easier to control and have less concerns about the quality of signal flowing through them (unlike AUDIO or Control Voltage, which we need to be really precise!).

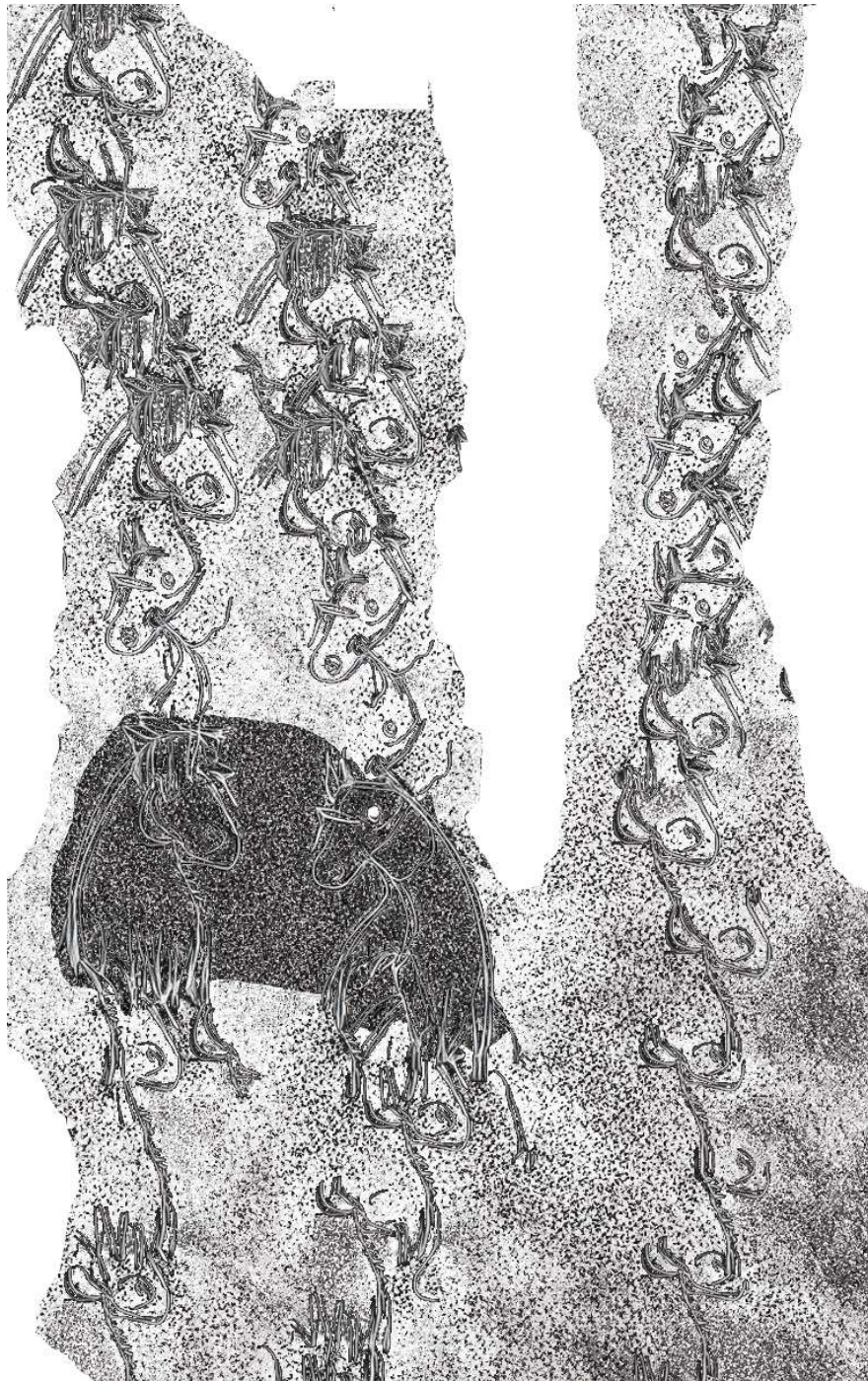
They can be HIGH for a long time, or just for a moment (sometimes called a TRIGGER). If a GATE signal is HIGH, you can think of the GATE as being "open", which allows

some musical process or sound to flow. When it is closed, that signal is used to halt or dampen a musical process.

For a sustained GATE signal, you can think of a musical note. The GATE goes HIGH, and the sound begins, and is held until the GATE closes (goes LOW). That's really common.

A really common way that GATE signal is used is as a TRIGGER, which goes on momentarily and then dips back to closed. This can also be used to cue a musical event like a note or indicate to another piece of hardware that a sequence has finished.

In the ESTUARY system here today, notice that CLOCK is common for GATE ins and outs. These are timed pulses that indicate tempo (think of musical beats). This allows us to synchronize musical meter (rhythmic structure) across different and independent hardware modules.



Percussion

(Wizard)

Our percussion module (which *should* be on the wizard faceplate) is our central timekeeper and rhythmic backbone. The percussion module has three musical voices: a kick drum, a hi-hat, and a snare drum.

Unlike a drum machine, you cannot specify exactly when certain events happen. Using the dials, you can modulate underlying algorithmic processes that have different rhythmic effects. For the trade off of precise control, you get musical effects quickly.

Dials 1 and 5 control speed and time “wavyness”. Gate out 1 should be patched to Rhythm Alien to sync time.

The kickdrum can be changed with dials 2 and 6 and switch 1. Switch 1 changes the base synthesis module (different sound). Dials 2 and 6 control parameters of what’s called a Euclidean sequencer -- basically you are changing the numerator and denominator of a fraction to generate different rhythms. You can patch in a “pitch out” control voltage to CV in 3 to control the pitch of the drum.

The snare and hi-hat work similarly. Switch 2 toggles snare sounds. The underlying algorithm is a Perlin noise grid. Dials 7 and 8 offset the grid to give you a different pattern. Change the on/off threshold with dials 3 and 4 to get more event density.

Percussion (Wizard)

Dials

- 1: BPM
- 2: kick pattern fill
- 3: hi hat density
- 4: snare density

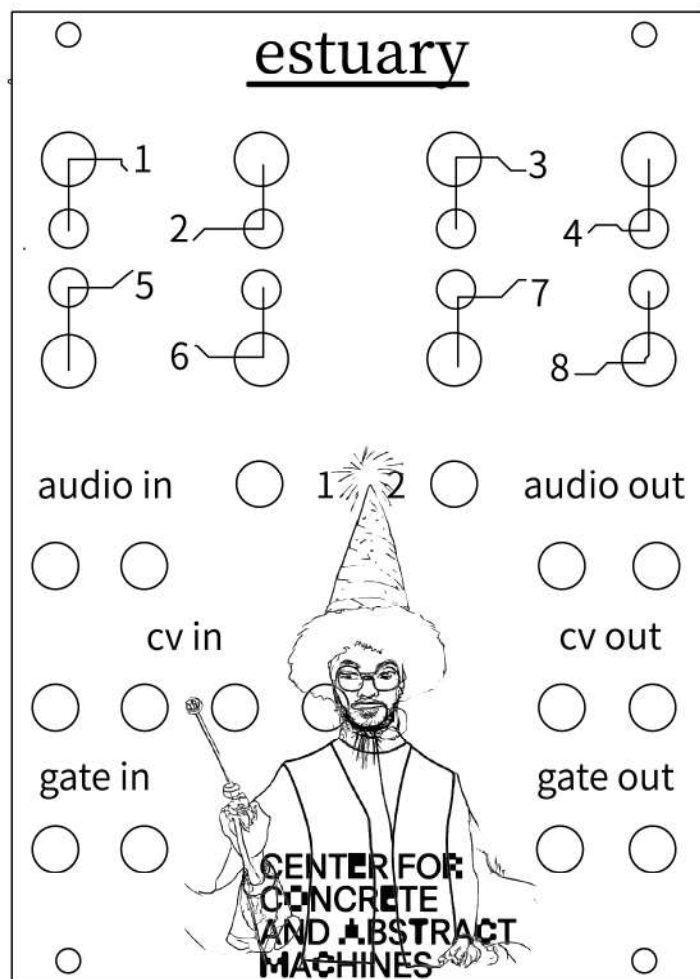
- 5: chaotic signal
- 6: kick pattern length
- 7: hi hat pattern offset
- 8: snare pattern offset

Switches

- 1: kick sound
- 2: snare sound

LEDs

- sequencer beat



Percussion (Wizard)

CV in

- 1: hi hat pan
- 2: snare pitch
- 3: kick pitch

Gate in

- 1: clock
- 2: hi hat open

CV out

- 1: perlin noise (hi hat)
- 2: chaotic signal (BPM)

Gate out

- 1: clock out
- 2: kick triggers

Rhythm (Alien)

Our rhythm module is keyboard player of the ensemble. It is a three-voice synth that changes voices according to an underlying shift-register (a technique typical in generative synthesis).

If no CLOCK signal is patched in to GATE 1, it won't go! So start there first. Next, try to patch PERCUSSION gate out 2 to RHYTHM GATE IN 1 (sync voice). Turn Dial 5 to extremes to see how it reacts; it either syncs with the kick drum or fills in when it's not playing.

Dial 6 and 7 control the probability of *if* notes change and the intervallic range of their change, respectively. Try patching CV out 1 or 2 to PERCUSSION's pitch CV ins.

Dials 1-4 and 8 control aspects of the sound's timbre. Attack is the speed at which a note gets to full volume; Decay is the speed at which it backs off. Switch 1 toggles between different waveforms.

Filter resonance and frequency (FQ) will change how a filter modifies waveform -- its harmonic quality (how much low and high sound we here). The attack of the filter changes the onset of that filter.

Dials

- 1: Attack (note start)
- 2: Decay (note end)
- 3: Filter resonance
- 4: Filter FQ (pitch)
- 5: Sync $\leftarrow \rightarrow$ Hocket
- 6: Voice chance prob
- 7: Voice jump range
- 8: Filter envelope attack

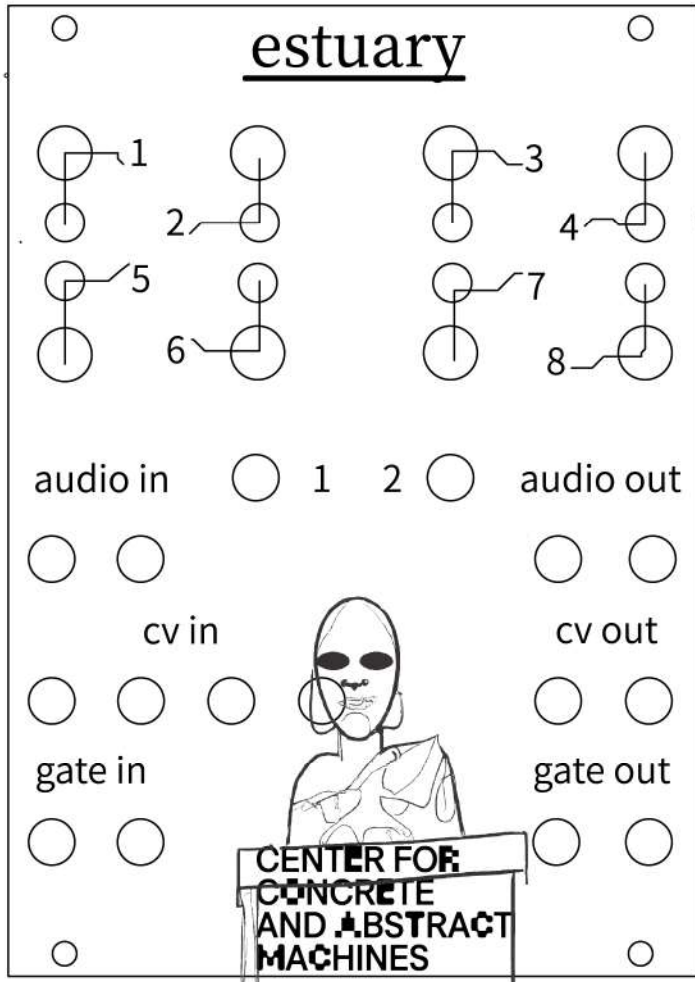
Switches

- 1: Waveform
- 2: Filter

LEDs

sequencer beat

Rhythm (Alien)



CV in

- 1: filter resonance

Gate in

- 1: clock
- 2: sync voice

CV out

- 1: top voice pitch
- 2: bottom voice pitch

Gate out

- 1: trigger on active
- 2: clock thru

Bass

((Cyberpotato))

BASS is a single-voice harmonic oscillator and synth voice. Like RHYTHM, it creates melodies using a shift-register; the LEDs display the activity of the shift register. Dial 5 controls how much the melody changes over time.

The remaining dials control aspects of the sound quality. The BASS module can either fill out the low end of your patch or can be turned into more of a lead melodic voice by adjusting the octave (dial 7) and the subbass voice (dial 8).

Dial 6 will create fluid movement between notes, a charming effect in moderation or a good way to signal a coming musical transition in excess.

Try patching CV OUTs 1 or 2 to kick drum patch.

Like RHYTHM, it will need a clock signal into GATE IN 1. If you've patched CLOCK from PERCUSSION to RHYTHM, RHYTHM's GATE OUT 2 will provide a CLOCK signal. While you're at it, you can take the GATE OUT 1 from RHYTHM and patch it to GATE IN 2 to provide accents to the bass module from Rhythm's activity.

Dials

- 1: Attack (note start) 5: Rand $\leftarrow \rightarrow$ Loop
2: Decay (note end) 6: Glide
3: Harmonics 7: Octave
4: Filter FQ (pitch) 8: Subbass

Switches

- 1: Write Pattern
2: Filter

LEDs

(shift register)

2-4pm // 04.17.26 @
Ezkenazi Museum
of Art; Indiana
University
Bloomington.

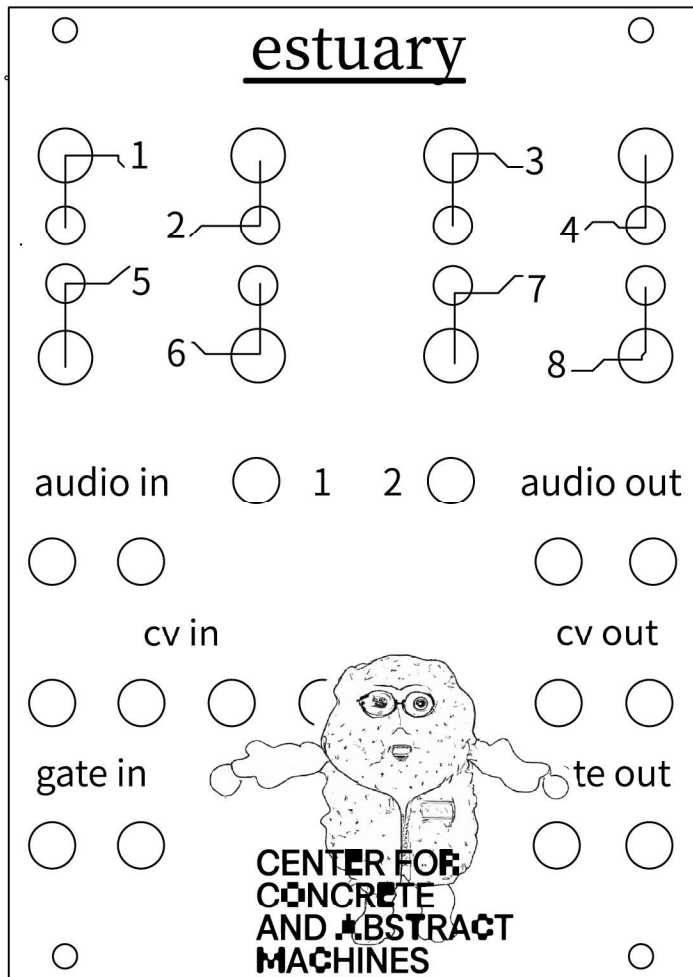
Zine illustrations,
writing, hardware,
and software
by Grace Grace
Grace.

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website: <https://ccam.world>.
instagram: @ccam_
chicago

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Bass (Cyber Nugget)



Bass (Cyber Nugget)

CV in

1: filter
resonance

Gate in

1: clock
2: accent

CV out

1: top byte
pitch
2: bottom byte
pitch

Gate out

1: accent
2: clock thru

CENTER FOR
CONCRETE
AND ABSTRACT
MACHINES